All Your BI

Senior AI Engineer

ASSIGNMENT · DEEP-DIVE STAGE

Thanks for making it to the assignment stage. This assignment helps us understand your strategic thinking, how you solve problems and how you break down ambiguous challenges. We're looking for your reasoning, not perfection, so feel free to share your thought process as you work through it.

How this works

→ Read both scenarios below and choose one to prepare. Pick whichever genuinely interests you most.

→ You'll have 3 business days to prepare and submit your solution.

→ Your deliverable is a presentation (slides or a shared document) that you'll present during the deep-dive interview. Plan for around 15 minutes to walk us through your thinking and your approach to the four questions in your chosen scenario.

→ Use whichever presentation format you're most comfortable with. Slides, a shared document or a similar format all work well.

Tools and AI

Use whatever you'd normally reach for to prepare, AI tools included. Honestly, we'd expect a Senior AI Engineer candidate to use AI well while prepping for this. Just be ready to say briefly how you used it.

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Pick one scenario

SCENARIO 1

Data-quality investigation copilot

We have a deterministic data-quality engine: rules run against a terminal's data warehouse and emit PASS / WARN / FAIL with evidence. When something WARNs or FAILs, a human traces the root cause back through the pipeline (query layer, staging, Data Vault, reporting marts) and writes up symptoms, root cause and resolution. We keep a small JSON "case library" of past investigations as memory.

We want an LLM agent to do the first-pass investigation and propose a root cause, so the engineer only reviews.

QUESTIONS

1. How would you stop it from confidently inventing root causes?
2. The case library and the evidence it reads include free text written by engineers and pulled from pull-request comments and ticket fields. Any concerns with feeding that to the agent? How would you handle them?
3. How do you know it's actually good, and that it stays good as models and prompts change over time?
4. Which parts of this should an LLM not touch?

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SCENARIO 2

“Ask your terminal data” (client-facing)

A client wants to ask plain-English questions of their terminal data, for example “which berths had the worst crane idle time last month?” or “why did dwell time spike for reefer containers in week 12?”. The data is messy: inconsistent naming across terminals, late-arriving and corrected records, soft-deletes, and the same concept modeled differently per client. We want this to be a repeatable product across many clients, not a one-off.

QUESTIONS

1. Walk us through the architecture. How do you approach turning a question into a trustworthy answer?
2. A wrong answer could mislead a client’s operational decision. How do you make it trustworthy?
3. How do you keep it affordable and fast across many clients and many questions?
4. How would you know it’s genuinely useful to the client, not just an impressive demo?

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